Project Brief

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Book: Project Brief
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1. Project

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Module Code:	ITDCA0-22
Module Name:	Database Design Concepts
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Submission Week:	Week 6

This module is presented on NQF level 5.

5% will be deducted from the student's assignment mark for each calendar day the assignment is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Assignments submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is an individual project.

This project contributes 30% towards the final mark.

[1] Under no circumstances will assignments be accepted for marking after the assignments of other students have been marked and returned to the students.

2. Instructions

1. All work, including draft notes, must be submitted with the completed project.
2. The program must be operational with as few faults as possible.
3. 10% will be deducted from this project if it is returned for resubmission due to plagiarism.

Plagiarism and Referencing

Consult the section at the end of this document, which outlines how negative marking will be applied as well as the way in which it will affect the assignment mark.

3. Question 1

Question 1 [30 Marks]

Study the scenario and answer the questions that follow:

Conceptual Data Modelling

A new company has developed a Massive Multiplayer Online Role Playing Game (MMORPG). Players register accounts using a unique account name and pay a monthly subscription fee. Accounts become inactive when payment expires.

Each account can create multiple characters, each with:

- Name, team, and skill level
- Ability to collect items

Items are stored in an inventory with:

- Exactly 8 slots
- Stacking allowed (quantity per slot)

Errors are logged with types/categories.

1.1 Identify all entities and draw a high-level ERD (conceptual diagram) showing only entities and relationships (no attributes yet). (6 Marks)

1.2 Extend your ERD by adding attributes and unique identifiers (UIDs #) to each entity. (6 Marks)

1.3 Update your ERD to include:

- Cardinality (1:M, M:N)
- Optionality
- Relationship names

1.4 Your ERD currently contains a many-to-many relationship between CHARACTER and ITEM: (8 Marks)

- Redraw the ERD to resolve this using an intersection entity
- Clearly show the composite UID

(5 Marks)

1.5 Modify your ERD to enforce the rule that an inventory has exactly 8 slots.
Explain how this constraint is represented in your diagram. (5 Marks)

End of Question 1

4. Question 2

Question 2 [20 Marks]

Study the scenario and answer the questions that follow:

The ERD from Question 1 must now be implemented in a relational database.

2.1 Convert your ERD into a relational schema diagram (table-based diagram). Include: (10 Marks)

- Table names
- Primary Keys (#)
- Foreign Keys (labelled FK1, FK2, etc.)

2.2 Using your diagram, show how the intersection entity (inventory/selection) is mapped to a table. Include the composite Primary Key. (4 Marks)

2.3 Redraw part of your diagram to show how a recursive relationship could be added (e.g., characters forming teams with leaders). (3 Marks)

2.4 Indicate on your diagram where a Unique Key (UK) exists and explain why it is not the Primary Key. (3 Marks)

End of Question 2

5. Question 3

Question 3 [30 Marks]

Study the scenario and answer the questions that follow:

> **Normalisation with Diagram Transformation**
>
> You are given an unnormalised structure (0NF):
>
> **ACCOUNT_DATA**
> (accountName, subscriptionStatus, characterName, team, skillLevel, itemName, quantity, errorMessage, errorType)

3.1 Represent the above structure as a 0NF diagram (single entity with all attributes). Clearly indicate the chosen Primary Key (PK). (5 Marks)

3.2 Transform the diagram into 1NF, showing: (8 Marks)

- Separate entities
- Foreign Key relationships

3.3 Transform your diagram into 2NF, removing partial dependencies. (7 Marks)

3.4 Transform your diagram into 3NF, ensuring no transitive dependencies remain. (5 Marks)

3.5 Draw the final ERD derived from 3NF, showing all entities and relationships clearly. (5 Marks)

End of Question 3

6. Question 4

Question 4 [20 Marks]

Study the scenario and answer the questions that follow:

Advanced ERD Modelling

During system expansion, the following requirements are introduced:

- Items must now be categorised into Weapon, Armour, and Potion
- A character may only interact with one item type at a time
- Subscription history must be stored over time
- Errors must be analysed more efficiently

4.1 Redraw part of the ERD to include a supertype/subtype structure for ITEM. Indicate: (5 Marks)

- Supertype
- Subtypes
- Inheritance

4.2 Modify your ERD to include an arc (exclusive relationship) where a character can only use one subtype at a time. (5 Marks)

4.3 Extend your ERD to include a historical entity for subscription payments. (5 Marks)

4.4 Identify a potential fan trap or chasm trap in your diagram and redraw the corrected ERD portion. (5 Marks)

End of Question 4